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Iron loss of paddy soil in China and its environmental implications

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Abstract Iron (Fe) is an important element for the terrestrial and marine ecosystems through its biogeochemical cycling on the Earth's surface. China has a long rice cultivation history, with extensive rice distribution across many types of paddy soils. Paddy soils are the largest anthropogenic wetlands on earth with critical roles in ecosystem functions. The periodic artificial submergence and drainage during paddy soil evolution result in significant changes in soil moisture regime and redox conditions from the natural soils, which facilitate the increase of Fe solubility and mobilization. However, there is a lack of systematic assessment on the magnitude of the migration and loss amount of Fe from paddy soils. In order to quantify the Fe loss and assess the dynamic evolution of Fe in the soils after rice cultivation, seven paddy soil chronosequences derived from different landscapes (bog, plain, terrace) and parent materials (acidic, neutral, calcareous) with cultivation history from 0 to 2,000 yr were studied. Results showed that the rates and trajectories of Fe evolution showed distinct patterns among the studied seven paddy soil chronosequences. However, net losses of Fe from 1 m soil depth occurred at all studied paddy soil chronosequences regardless of the original landscapes and parent materials. Fe in the paddy soils derived from the calcareous lacustrine sediments in the bog area showed a slight accumulation during the initial stage (50 yr) of paddy cultivation, with a loss rate of 0.026 kg m^{-2} yr^{-1} during the 50- to 500-yr time period. For the paddy soils developed on the calcareous marine sediments in the plain area, Fe evolution was dominated by the internal movement in soil profiles through coupled reducing-eluviation reactions in the surface horizons and oxidation-illuviation in the subsurface horizons within 1,000 yr of paddy cultivation, with an averaged net loss rate of 0.029 kg m^{-2} yr^{-1} during the 1,000- to 2,000-yr time period of rice cultivation. In contrast, Fe in the paddy soils derived from the acidic and neutral parent materials in the plain and terraced upland areas was rapidly lost during the initial stage of paddy cultivation, with a maximum loss rate of 1.106 kg m^{-2} yr^{-1}, while the Fe loss rate decreased gradually with increasing paddy cultivation age. Soil pH, $CaCO_3$, and organic matter contents of the original soils, the length of time of paddy cultivation, landscape types and positions, and changes in soil moisture regime and redox condition induced by artificial submergence and drainage were the main factors controlling the rates and trajectories of Fe loss during paddy soils evolution. The amount of Fe loss caused by rice cultivation at the national scale was estimated based on the data collected from this study and the literature. The Fe loss fluxes of paddy soils in China were about 46.4–195.7 Tg yr^{-1}, and the amounts of Fe losses from paddy fields nationwide were about 5,121.5–9,412.2 Tg. Quantifying Fe loss from paddy fields is important to scientifically assess the impact of paddy cultivation on the Fe biogeochemical cycle.

Keywords Soil-water interface, Land-sea interface, Paddy soil, Soil Fe loss, Biogeochemical cycle

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1. Introduction

Iron (Fe) is an essential nutrient for life as it is indispensable to cell metabolic processes of animals, plants, and microorganisms. Taking the ocean as an example, in the high-nutrient and low-chlorophyll (HNLC) area, Fe is key in limiting the growth of plankton and thus primary productivity (Boyd et al., 2012; Liu et al., 2020). As such, many researches have proposed that adding Fe to the HNLC area may promote the growth of phytoplankton, accelerate the migration of carbon from the ocean surface to deep layers, reduce carbon dioxide in the atmosphere, and ultimately alleviate the greenhouse effect (Martin and Fitzwater, 1988; Martin, 1990; Song and Wang, 2020). As an important biogeochemical process, the Fe cycle in the earth surface system affects the terrestrial and marine ecosystems and the global climate. Jickells et al. (2005) reported the impact of Fe-containing dust transported from the land to the ocean through the atmosphere on the marine biogeochemical processes and the global climate. Lannuzel et al. (2007) found that the dissolution of sea ice was an important source of Fe in the surface water of Antarctica. Lam and Bishop (2008) proposed that the Fe input from the continental margin was an important source of Fe to the HNLC North Pacific Ocean. Li X Y et al. (2019) indicated that the labile Fe released by the melting of Asian glaciers had a significant impact on the global biogeochemical cycle of Fe.

Global Fe loss fluxes to the oceans through above-mentioned various transport pathways were very different due to the influence of climate, soil types, and delivery paths. Runoff through the land-sea interface is a major route of Fe transport into the oceans (Poulton and Raiswell, 2002). Anthropogenic activities and land use changes can reshape the soil environment and pedogenic processes (Zhang G L et al., 2019; Zhang et al., 2021) which have a significant impact on the migration pattern and intensity of soil Fe. Periodic artificial irrigation and drainage lead to strong soil-water interface processes in paddy soil systems, and the soil environment has been greatly changed from the original soil. This change has a great impact on the migration and transformation processes of C, N, P, S, Fe, which in turn produces significant environmental effects (Ponnamperuma, 1972). Fe is an element with variable valences; its reduction eluviation and oxidative illuviation is regarded as the characteristic soil forming processes of paddy soils (Li, 1992). The mechanisms of Fe leaching in paddy soils include mechanical leaching, chelation, and reductive eluviation which are closely related to the organic matter content, soil pH, and soil moisture regime and redox conditions (Yu, 1985). The solubility and mobility of soil Fe may be increased during the period submergence of rice cultivation. The dissolved Fe could be translocated within soil profile, or be leached out of the soil profiles during paddy soil evolution, thus participating in the regional or/and global biogeochemical Fe cycling. Therefore, quantifying the amount of Fe loss from paddy fields is critical for understanding the Fe cycle in the earth surface system. China is the biggest rice producer in the world with extensive rice distribution and long rice cultivation history which dates back to more than 10,000 yr ago (Gong et al., 2007; Gao et al., 2020). The area of paddy fields in China is currently about 4.57×10^7 ha (Liu et al., 2006), about 27% of that of the whole world (Liu et al., 2021). However, there is limited available information regarding long-term studies of the spatiotemporal dynamics of soil Fe loss after paddy cultivation at the global and regional scales. Such data are essential in evaluating the ecological impacts of paddy cultivation activities in China on the neighboring river or/and marine environment.

At the pedon scale, the Fe mobilization, translocation, redistribution, and the associated Fe mineral transformation are mainly determined by the soil moisture regime and initial parent material characteristics (e.g., pH, Fe concentration and compositions, weathering degree) (Yu, 1985; Huang et al., 2018a). Zhang and Gong (1993) compared the difference of Fe activation and migration under simulated submerged condition between the calcareous fluvo-aquic soil and the acid Quaternary red clay. Results showed that the Fe migration coefficient of the Quaternary red clay was much greater than that of the fluvo-aquic soil. Chen et al. (2011) showed that Fe migration in the paddy soils developed on the calcareous marine sediments was dominated by the internal cycles in soil profiles through coupled reducing-eluviation reactions in the surface horizons and oxidation-illuviation in the subsurface horizons during 1,000 yr of paddy cultivation. Huang et al. (2018a) revealed that total Fe concentration of the paddy soils derived from the acid Quaternary red clay in the subtropical humid Ferrisols areas decreased exponentially with increasing paddy cultivation age. Paddy soils in China originated from many types of original soils or different parent materials, and various landscapes (Gong, 1986). So, a scientific and comprehensive evaluation of the characteristic of Fe loss in paddy fields in China should consider both the original background and the length of time for paddy cultivation. In the current study, seven paddy soil chronosequences derived from different landscapes (bog, plain, terrace) and parent materials (acidic, neutral, calcareous) with cultivation history from 0 to 2,000 yr were studied and representative soil profiles data were collected according to Soil Series of China to investigate the spatiotemporal dynamics of soil Fe loss after paddy cultivation at the global and regional scales and to identify the important factors influencing Fe dynamics and possible mechanisms. These findings may be useful to better understand the global Fe biogeochemical cycle and its potential ecological effects on

the coastal and offshore marine environment.

2. Materials and methods

2.1 Study area

Paddy soil (Stagnic Anthrosols) refers to the soil with an anthrostagnic epipedon and a hydragric horizon formed under frequent artificial flooding and fertilization. It is an important soil resource in China. Paddy soil is found in almost all regions of the country, from Huma of Heilongjiang Province to Sanya of Hainan Province, from Taiwan Province to the Yili River Valley and Kashgar region of Xinjiang, from plateaus at an altitude of 2,700 m to lowlands below the sea level (Li, 1992; Gong et al., 2014). About 93% of the paddy soil is distributed in the Yangtze River Basin and southern China, that is south of the Qinling-Huaihe Line and east of the Qinghai-Tibetan Plateau. Only 7% of the paddy soil is distributed in northern China. Although there has been an increase in paddy fields in Northeast China recently, the vast majority of paddy soil is still located in subtropical regions (Figure 1). Since both the original background of paddy soil and the age of rice cultivation need to be considered, seven typical rice-producing areas across China were selected in this study based on the temporal and spatial distribution of ancient rice in China (Gong et al., 2007) (Figure 1). The specific information of the seven study sites from north to south is listed in detail as follows. (1) Jiangsu Baoying bog soil area (B-C-BY), located on the northern edge of the northern subtropical zone, with a mean annual temperature (MAT) of 15.0°C and a mean annual precipitation (MAP) of 1,042 mm. The area is a lagoon-type depression, with a low-lying area of land surrounded by high terrain. Soil parent materials were mainly lacustrine sediments with some river alluvium, with some $CaCO_3$ content, high soil organic carbon (SOC), and a pH greater than 7 (Table 1). (2) Lower Yangtze Plain area (P-C-CX), located on the southern edge of the northern subtropical zone, with a MAT of 16.3°C and a MAP of 1,325 mm. This area is a marine plain. The parent materials were mainly coastal sediments, with a pH greater than 8 and a relatively higher amount of $CaCO_3$. (3) Guanghan mountain and hilly region in Sichuan (T-N-GH), located in the mid-subtropical zone, with a MAT of 17.8°C and a MAP of 890 mm. The parent materials were Quaternary sediments with a pH close to 7. (4) Yingtan low hills area in Jiangxi (T-A-YT), located in the mid-subtropical zone, with a MAT of 17.6°C and a MAP of 1,795 mm. The parent materials were weathered red sandstone, with a pH of around

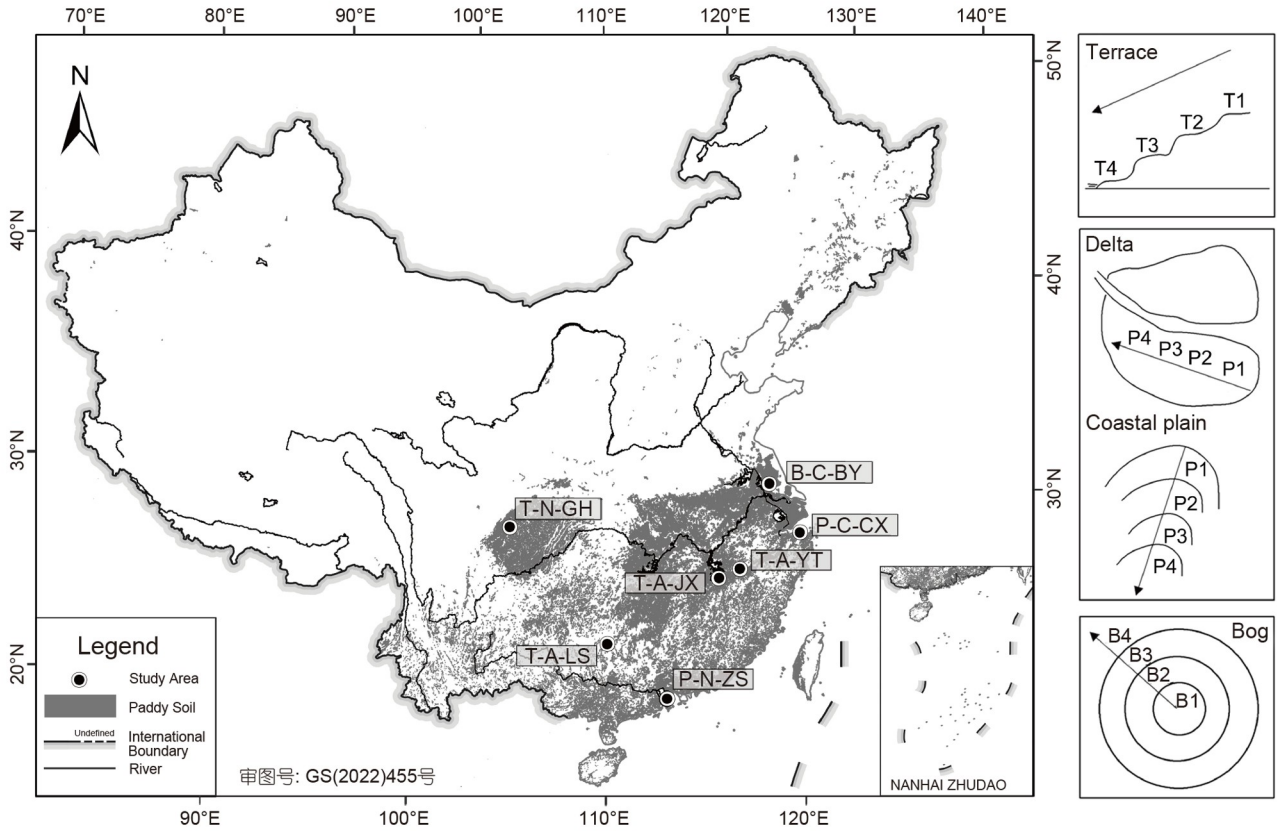


Figure 1 Distribution of paddy soil in China, the position and spatial layout of the studied seven soil chronosequences, and the relationship of landscape characteristics and paddy cultivation age. In the boxes on the righthand side, T denotes terraces, P denotes plains, B denotes bogs, and arrow direction indicates the increase of paddy cultivation age.

Table 1 Description and classification of the studied profiles

Profile code[1)]	Paddy cultivation age (yr)	Soil type[2)]	Landuse	Horizon[3)]	pH[4)]	$CaCO_3$(%)[4)]	SOC(%)[4)]
B-C-BY, Baoying, Jiangsu, northern edge of northern subtropical zone, 15.0°C, 1,042 mm, bog, calcareous lacustrine sediments, Zhang and Gong (2003)[5)]							
B-C-BY01	0	Gleysols	Bog	Ah-A-G	7.28	1.54	5.07
B-C-BY02	5	Gleysols	Rice-wheat	Ag-BG1-BG2-G	7.49	1.06	6.60
B-C-BY03	20	Cambisols	Rice-wheat	Ap-Bg1-Bg2-BG-G	7.49	1.03	2.28
B-C-BY04	50	Stagnic Anthrosols	Rice-wheat	Ap1-Ap2-Br-Bg2-Bg3-BG	7.69	0	2.10
B-C-BY05	500	Stagnic Anthrosols	Rice-wheat	Ap1-Ap2- BE-E-Br	6.67	0	3.04
P-C-CX, Cixi, Zhejiang, southern edge of northern subtropical zone, 16.3°C, 1,325 mm, marine plain, calcareous coastal sediments, Chen et al. (2011)							
P-C-CX01	0	Entisols	Uncultivated mudflat	C1-C2-C3-C4	8.07	5.83	1.04
P-C-CX02	50	Stagnic Anthrosols	Single rice	Ap1-Ap2-Bg1-Bg2-Bg3-BCg	8.13	2.32	0.67
P-C-CX03	300	Stagnic Anthrosols	Single rice	Ap1-Ap2-Bg1-Bg2-Bg3-BCg	8.13	1.88	0.71
P-C-CX04	700	Stagnic Anthrosols	Single rice	Ap1-Ap2-Eg-Btg1-Btg2-Ab-Bb	7.33	0	0.73
P-C-CX05	1000	Stagnic Anthrosols	Single rice	Ap1-Ap2-Btg1-Btg2-Btg3-Ab-Bb	7.49	0	0.63
P-C-CX06	2000	Stagnic Anthrosols	Single rice	Ap1-Ap2-Bg1-Bg2-Bg3-Ab-Bb	5.99	0	0.93
P-N-ZS, Zhongshan, Guangdong, southern sub-tropical zone, 22.0°C, 2,042 mm, alluvial plain, neutral river alluvium, Zhang and Gong (2003)							
P-N-ZS01	0	Cambisols	Banana grove	A1-A2-Bg1-Bg2-G	7.7	0	1.69
P-N-ZS02	50	Cambisols	Double rice	Ap-AB- B1-B2-BG-G	7.27	0	1.82
P-N-ZS03	200	Stagnic Anthrosols	Double rice	Ap1-Ap2-Bg1-Bg2-BG	7.23	0	1.58
P-N-ZS04	600	Stagnic Anthrosols	Double rice	Ap1-Ap2-Bg1-Bg2-Bg3-G	6.59	0	2.41
T-N-GH, Guanghan, Sichuan, mid subtropical zone, 17.8°C, 890 mm, terrace, neutral Quaternary sediments, this study							
T-N-GH01	0	Cambisols	Waste upland	A1-A2-B-BC-C	6.78	0	0.68
T-N-GH02	100	Stagnic Anthrosols	Single rice	Ap1-Ap2-Br1- Br2-C	6.87	0	0.67
T-N-GH03	1000	Stagnic Anthrosols	Single rice	Ap1-Ap2-Btr1-Btr2-C	7.21	0	0.81
T-A-YT, Yingtan, Jiangxi, mid subtropical zone, 17.6°C, 1,795 mm, terrace in low mountain and hilly area, acid weathered red sandstone, Han and Zhang (2013)							
T-A-YT01	0	Cambisols	Waste upland	A-B-BC	4.87	0	0.27
T-A-YT02	30	Cambisols	Rape-rice	A-B-BC	5.49	0	0.44
T-A-YT03	100	Stagnic Anthrosols	Rape-rice	Ap1-Ap2-Br-BC	5.53	0	0.30
T-A-YT04	200	Stagnic Anthrosols	Rape-rice	Ap1-Ap2-Br-BC	5.48	0	0.49
T-A-JX, Jinxian, Jiangxi, mid subtropical zone, 17.3°C, 1,549 mm, terrace in low mountain and hilly area, acid Quaternary red clay, Han and Zhang (2013)							
T-A-JX01	0	Ferrisols	Waste upland	A-Bt1-Bt2-BC	4.63	0	0.41
T-A-JX02	60	Stagnic Anthrosols	Rape-rice	Ap1-Ap2-Br1-Br2-BC	5.14	0	0.74
T-A-JX03	150	Stagnic Anthrosols	Rape-rice	Ap1-Ap2-Br1-Br2-BC	6.37	0	0.64
T-A-JX04	300	Stagnic Anthrosols	Rape-rice	Ap1-Ap2-Br1-Br2-BC	6.13	0	0.69
T-A-LS, Longsheng, Guangxi, subtropical zone, 18.2°C, 1,557 mm, terrace in high mountain and hilly area, acid weathered purple sandy shale, Han and Zhang (2013)							
T-A-LS01	0	Cambisols	Waste upland	A-B-BC	5.37	0	0.38
T-A-LS02	30	Cambisols	Double rice	Ap-AB-Bw-B-BC	5.21	0	0.38
T-A-LS03	150	Stagnic Anthrosols	Double rice	Ap1-Ap2-Bg-Br1-Br2-BC	4.88	0	1.23
T-A-LS04	300	Stagnic Anthrosols	Double rice	Ap1-Ap2-Bg-BC	4.77	0	1.33

1) The profile code includes three parts. The first part is the landscape category, including B(bog), P(plain), and T(terrace). The second part is the parent material, including C(calcareous), N(neutral), and A(acidic). The third part is the study area, for example, BY(Baoying), CX(Cixi), ZS(Zhongshan), GH (Guanghan), YT(Yingtan), JX(Jinxian), and LS(Longshen). The same applies below. 2) According to the Chinese Soil Taxonomy (Gong, 1999). 3) The capital letters A, E, B, G, and C represent the genetic horizons in soils. For horizons (transitional horizons) dominated by properties of one master horizon but having subordinate properties of another, two capital letter symbols are used, such as AB, BE, BG, and BC. Lower case letters are used as suffixes denote characteristics of genetic horizons. 4) Average value of the profile. 5) Contents in this row include profile code, sampling site, climate zone, mean annual temperature, mean annual precipitation, terrain/landscape, parent material, and reference

5 and a low SOC content. (5) Jiangxi Jinxian low hill red soil area (T-A-JX), located in the mid-subtropical zone, with a MAT of 17.3°C and a MAP of 1,549 mm. The parent materials were Quaternary red clay, with a pH of around 5 and a low SOC content. (6) Guangxi Longsheng alpine and hilly area (T-A-LS), located in the subtropical zone, with a MAT of 18.2°C and a MAP of 1,557 mm. The parent materials were weathered purple sand shale, with a pH close to 5. (7) The Pearl River Delta rice-producing area (P-N-ZS), located in the South Asia tropical zone, with a MAT of 22.0°C and a MAP of 2,042 mm. The parent materials were mainly river alluvium, and the pH is slightly greater than 7 with no lime reaction. The original landscape of the paddy fields in the seven study areas can be divided into 3 categories according to the characteristics of soil moisture regime, namely, poorly drained bog areas with a near surface water table (B-C-BY), alluvial and marine plains with groundwater fluctuation (P-C-CX, P-N-ZS), and terraces with relatively good drainage (T-N-GH, T-A-YT, T-A-JX, T-A-LS). The parent materials can also be divided into 3 categories according to the pH and $CaCO_3$ content, namely, calcareous (B-C-BY, P-C-CX), neutral (P-N-ZS, T-N-GH), and acidic (T-A-YT, T-A-JX, T-A-LS) (Table 1).

2.2 Paddy soil chronosequences recognition

A soil chronosequence is defined as a genetically related sequence of soils developed on similar parent materials and relief under the influence of constant or ineffectively varying climatic and biotic factors. The principle of chronosequence approach is a space-for-time substitution with soils that differ only with respect to time (soil age) while assuming that all of the other soil-forming factors remain constant (Chen and Zhang, 2011). The age of the soils in this study refers to the time length of artificial rice cultivation, which was estimated based on comprehensive information, including relationship between landscape characteristics and paddy cultivation time (Figure 1), local historical literature (county chronicles, water conservancy chronicles, and village development history), and on-site investigations. For anthropogenic soils, landscape features are closely related to the cultivation time. A bog landscape usually has a low-lying middle area with high surroundings, and the distribution of soils follows a concentric pattern with a gradually shortened temporal length of rice cultivation from outside to inside (Zhang and Gong, 2003). The paddy soil chronosequence (B-C-BY) consisting of four profiles with approximately 5, 20, 50, and 500 yr paddy cultivation history was determined based on landscape characteristics, Baoying county chronicles, and field investigations (Table 1). In addition, an uncultivated bog soil profile was selected to represent the original soil (parent material, time zero) of the paddy soils. The cultivation age of the paddy soils derived from marine plain landscape was mainly determined based on the direction of the coastline extension and the chronology of dyke construction (Chen et al., 2011). The paddy soil chronosequence (P-C-CX) consisting of five profiles with approximately 50, 300, 700, 1,000, and 2,000 yr of paddy cultivation history, and an uncultivated mud beach profile representing the original soil (parent material, time zero) of the paddy soils were identified. The paddy soil chronosequence (P-N-ZS) consisting of three profiles with approximately 50, 200, and 600 yr paddy cultivation history was determined based on the evolution direction of the alluvial fan, local county chronicles, water conservancy records, and other related historical documents (Zhang and Gong, 2003). In addition, a Cambisols profile was selected to represent the original soil (parent material, time zero) of the paddy soils. For terraced landscapes, soils at the bottom of slopes were generally the first to be converted to paddy field due to better moisture conditions, thicker soil layers, and proximity to the village; as population pressure increased, lands upslope were progressively brought into paddy cultivation. Thus, these hillside terraces, with increasing cultivation age from the top to the bottom of the slopes, provide soil chronosequences (Han and Zhang, 2013). The history of paddy cultivation in the older profile at the bottom of the slope was determined from local historical literature from when the nearby villages were settled. For the newer profile at the top of the slope, information was obtained by interviewing the local farmers. The uncultivated soil profile at the highest slope position was treated as the original soil, i.e., time zero with respect to paddy cultivation. The paddy cultivation age of the four paddy soil chronosequences (T-N-GH, T-A-YT, T-A-JX, T-A-LS) in terrace landscapes in this study was determined based on above-mentioned approach (Table 1). Note that the soil paddy cultivation ages marked in this study are not absolute, but the relative order of paddy cultivation ages in each chronosequence is reliable.

2.3 Field sampling and laboratory analyses

Within each area of identical paddy cultivation history, one representative profile was chosen for soil sampling based on the soil landscape and morphological characteristics of that area. All soil samples were collected when the fields were drained after rice harvest. Soil profiles were described and sampled according to genetic horizons following standard field description guidelines (Schoeneberger et al., 2012). The detailed field descriptions and classifications of the studied profiles are given in Table 1.

The collected soil samples were dried at room temperature and then gently crushed using a wooden pestle and mortar and passed through a 2-mm nylon sieve. Soil bulk density was measured on the 100 cm^{-3} undisturbed soil cores by drying the cores for 24 h at 105°C. Soil pH was determined

at a 1:2.5 soil/solution ratio using distilled water and the carbonate content was determined using a Dietrich Fruhling pressure calcimeter. Soil organic carbon (SOC) was measured by the Walkley-Black wet oxidation method using the 149-μm fraction. For total Fe analysis, soil samples (<74 μm) were fused by a mixture of 1:1 lithium metaborate and lithium tetraborate for 30 min in a 1,000°C muffle furnace and then were dissolved in 10% HNO_3+1% HF solution. Total Fe concentration was determined by inductively coupled plasma-optical emission spectrometry (Zhang and Gong, 2011). We estimated the precision as 5–10% relative standard deviation based on replicates and standard samples (Geochemical Standard Reference Sample Soil, GSS-3, GSS-5, and GSS-8). Fe mass (kg m^{-2}) in the soil pedon (0–100 cm) (Zheng et al., 2013) was calculated by multiplying Fe concentration by bulk density and thickness of soil horizon using the following equation:

$$\mathrm{Fe}_{\mathrm{mass}} = \sum_{i}^{n} C_i D_i E_i / 100, \quad (1)$$

where, $\mathrm{Fe}_{\mathrm{mass}}$ is the Fe density (kg m^{-2}) in the soil pedon (0–100 cm), C_i, D_i, and E_i is, respectively, the Fe concentration (g kg^{-1}), bulk density (g cm^{-3}) and depth (cm) in the i horizon, and 100 is a unit conversion factor.

2.4 Data collection and processing

The coordinates, total Fe, bulk density, and layer thickness of the genetic horizons of 351 profiles in the "Soil Series of China" (Lu, 2017; Ma and Zhang, 2017; Wang, 2017; Zhang et al., 2017; Zhang and Ma, 2017; Wu K N et al., 2019), national soil parent material type data in vector format (converted to raster format in ArcGIS) (Xiong, 1986) and national land cover data with 100 m resolution (Zhang, 2007) were collected to estimate amounts of soil Fe losses induced by paddy cultivation at the national scale. The Fe density of 1 m depth of above-mentioned each profile was calculated using eq. (1). Then the spatial distribution map of paddy soils derived from different parent materials was extracted by overlaying the raster format of national soil parent materials map with the national 100 m land cover data in ArcGIS and eventually the total areas of paddy soils in different spatial locations with the same parent material was obtained using statistical analysis of ArcGIS. The paddy and non-paddy soil profiles developed from the same parent materials were extracted by overlaying the above extracted spatial distribution map of the paddy soil derived from different parent materials with the site distribution map of the 351 soil profiles obtained by importing the collected coordinates of these profiles into ArcGIS. Finally, the amounts of Fe losses caused by rice cultivation were calculated using the following equation.

$$\mathrm{Fe}_{\mathrm{loss}} = \sum_{i}^{n} (NP\mathrm{Fe}_i - P\mathrm{Fe}_i) PA_i / 10^6, \quad (2)$$

where $\mathrm{Fe}_{\mathrm{loss}}$ is the amount of Fe loss (Tg) caused by rice cultivation, $NP\mathrm{Fe}_i$, $P\mathrm{Fe}_i$, and PA_i is, respectively, the Fe density (t ha^{-1}) of non-paddy soil, Fe density (t ha^{-1}) of the paddy soil, the area of paddy soils (km^2) in the i parent material, and 10^6 is a unit conversion coefficient. The main errors of national-scale Fe loss of paddy fields estimated with this approach included (1) insufficient soil profiles, which could not cover all the parent material types of paddy soil in China, and (2) the heterogeneity of the same parent material in different regions and even in the same region.

3. Results

3.1 Chronosequential changes in total Fe concentration

The vertical distribution of total Fe concentration in seven uncultivated pedons was relatively uniform, compared with the paddy soils (Figure 2), showing that no perceptible pedological Fe changes had occurred. The maximum of total Fe concentration was found in the Quaternary red clay (Udic Ferrisols) located in the hilly slope landscape of Jinxian, Jiangxi, with a profile average of 65.8 g kg^{-1}. The minimum value was found in the lacustrine sediment (Gleysols) located in the Baoying bog landscape in Jiangsu, with a profile average of 32.1 g kg^{-1}. The higher total Fe concentration in T-A-JX01 than B-C-BY01 was attributed to the differences in parent materials. A measurable profile differentiation of total Fe was observed in all paddy soils (Figure 2), suggesting a transport and redistribution of Fe during paddy soil evolution. However, chronosequential changes in total Fe of the studied seven paddy soil chronosequences showed different dynamic patterns. In terms of dynamic change patterns, the studied chronosequences can generally be divided into four groups. The first group includes the paddy soils developed from calcareous lacustrine sediments in the bog landscape (Figure 2, B-C-BY). The total Fe in the surface horizon showed little change within 50 yr of rice cultivation, while the total Fe in the subsurface horizon gradually increased with increasing paddy cultivation age. However, the measurable decreases in total Fe concentration occurred within 0–70 cm depth after 50 yr of paddy cultivation. The second group includes the paddy soils derived from calcareous coastal sediments in the marine plain landscape (Figure 2, P-C-CX). The total Fe in paddy profiles showed a measurable sequential trend with time within 1,000 yr of rice cultivation. The differentiation of total Fe between surface and subsurface horizons was stronger with increasing paddy cultivation time, especially in those older soils (700- and 1,000-yr paddy cultivation history). The amount of total Fe in the illuvial horizons (Bg or Btg horizon) increased gradually with increasing rice cultivation age and the depth of illuviation is lowered gradually. However, the measurable

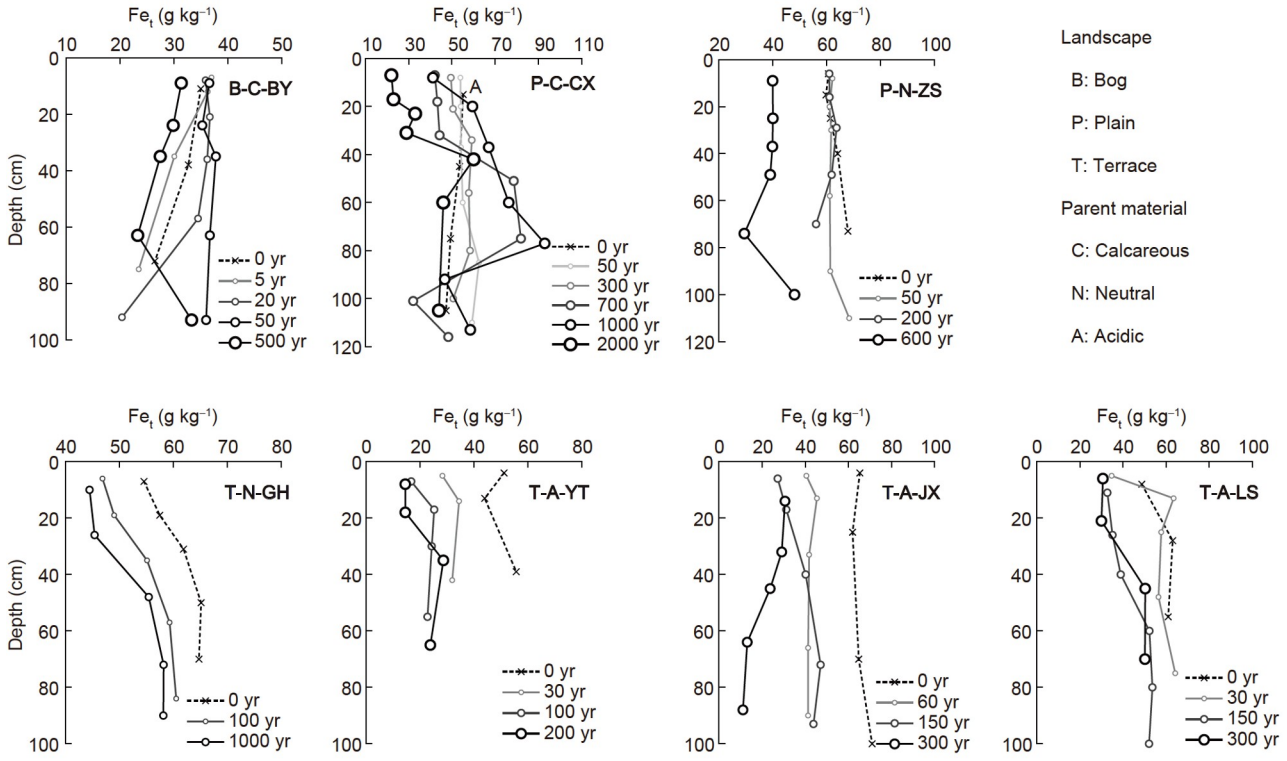


Figure 2 Distributions and variations of total Fe (Fe$_t$) concentrations along the soil depth and chronosequence.

decreases in total Fe concentration occurred throughout the profile after 1,000 yr of paddy cultivation. The third group includes the paddy soils originated from neutral river alluvium in the alluvial plain landscape (Figure 2, P-N-ZS). The total Fe in the upper 40 cm depth showed little change within 200 yr of rice cultivation, while the gradual decreases was found below the 40cm depths. However, the remarkable decreases in total Fe concentration occurred throughout the profile after 200 yr of rice cropping. The last group includes the paddy soils derived from neutral and acidic parent materials in terrace landscape (Figure 2, T-N-GH, T-A-YT, T-A-JX, T-A-LS). The total Fe concentrations in this group followed the trend that they were consistently lower in paddy soils than the uncultivated soil derived from fairly uniform parent materials. The total Fe for each paddy soil chronosequence in this group showed a gradual trend of depletion throughout the profiles with increasing paddy cultivation time.

3.2 Chronosequential changes in soil Fe density and Fe loss rate

Seven paddy soil chronosequences in this study can generally be divided into two groups according to the dynamic change patterns of Fe stocks and Fe loss rate. The first group includes the paddy soils developed from calcareous lacustrine sediments in the bog landscape (Figure 3, B-C-BY) and the paddy soils derived from calcareous coastal sediments in the marine plain landscape (Figure 3, P-C-CX). The dynamic change processes of Fe stocks can generally be divided into two contrasting stages for this group. The first stage (from the uncultivated soil to the 50-yr paddy soil in B-C-BY and from the uncultivated soil to the 1,000-yr paddy soil in P-C-CX) is characterized by the slight increases of Fe stocks. The second stage (during the 50–500-yr time period in B-C-BY and during the 1,000–2,000-yr time period in P-C-CX) is characterized by the measurable Fe loss. The second group includes the paddy soils derived from neutral river alluvium in the alluvial plain landscape (Figure 3, P-N-ZS) and neutral and acidic parent materials in terrace landscape (Figure 3, T-N-GH, T-A-YT, T-A-JX, T-A-LS). The Fe stocks in the paddy soils were measurably lower than in the uncultivated soils. The results suggest a profound anthropogenic impact on soil Fe loss. The Fe stocks decreased gradually with increasing paddy cultivation age in this group.

We calculated the Fe loss flux and loss rate in the seven chronosequences based on the total Fe density of the original soils and the paddy soils, and the paddy cultivation age (Table 2). The Fe loss fluxes ranged from −13.3 to 69.1 kg m^{-2}, the maximum Fe loss occurred in the 300 yr soil profile developed from the acid Quaternary red clay (T-A-JX) in the terraced landscape. Fe net loss in the paddy soil chronosequences developed from calcareous lacustrine sediments in bog landscape (Table 2, B-C-BY) and calcareous

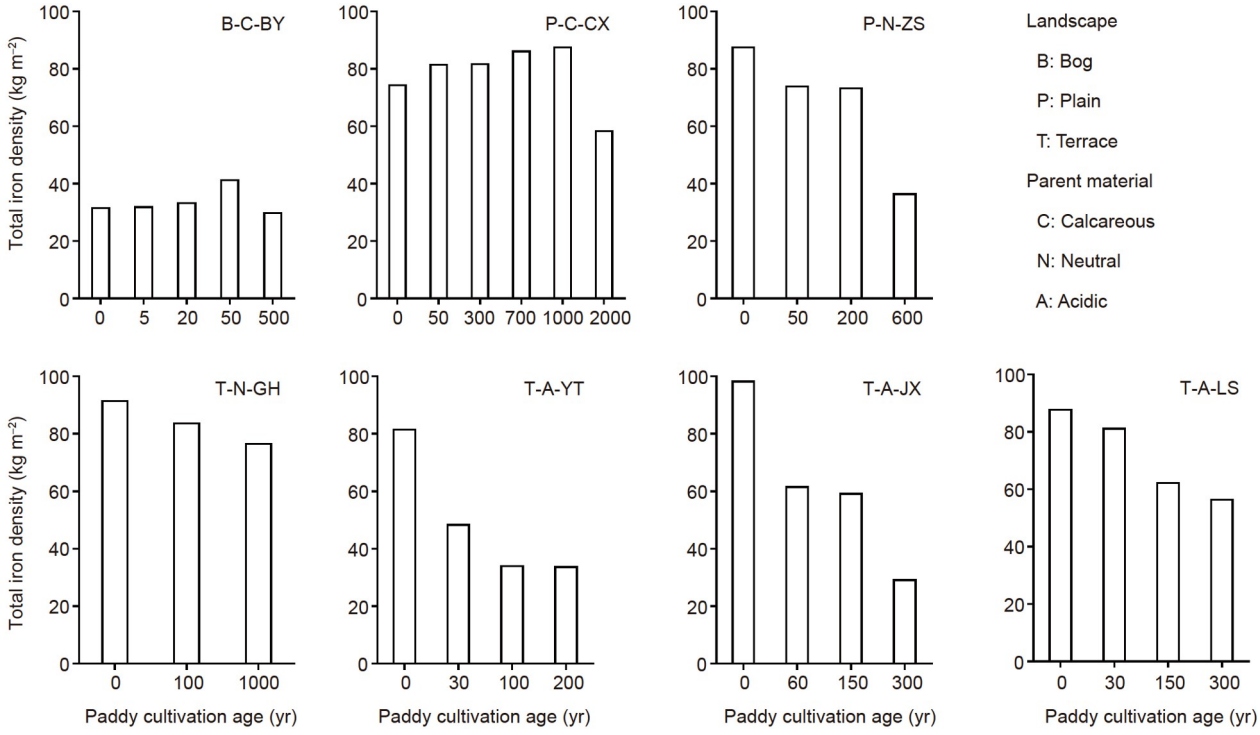


Figure 3 Changes in total Fe density within 1 m soil layer with paddy cultivation age.

coastal sediments in the marine plain landscape (Table 2, P-C-CX) occurred after 50 yr and 1,000 yr of rice cultivation, respectively. The average Fe loss rate was 0.026 kg m^{-2} yr^{-1} during the 50- to 500-yr time period of rice cropping in the B-C-BY, and 0.029 kg m^{-2} yr^{-1} during the 1,000- to 2,000-yr time period in the P-C-CX. The Fe loss flux in the remaining five paddy soil chronosequences (Table 2, P-N-ZS, T-N-GH, T-A-YT, T-A-JX, T-A-LS) increased gradually with increasing paddy cultivation age. Despite Fe loss flux showed an increasing trend with time, the Fe loss rate substantially decreased with the increasing rice cultivation age. Thus, the maximum (1.106 kg m^{-2} yr^{-1}) of Fe loss rate occurred in the initial stage of paddy cultivation (Table 2).

3.3 Soil Fe loss from paddy soils at the national scale

The total amount of soil Fe loss in paddy soils at the national scale was estimated using three methods. In the first method, the 28 soil profiles in the seven chronosequences studied were divided into two categories that are paddy soils and non-paddy soils (original soils). The average values of the total Fe density in the paddy soils (21 profiles) and the non-paddy soils (7 profiles) were 58.2 and 78.8 kg m^{-2}, respectively. The difference between the paddy and non-paddy soils could be considered as the net Fe losses caused by rice cultivation, which was 20.6 kg m^{-2}. The net Fe losses from paddy fields in China were then estimated to be 9,414.2 Tg according to the total paddy soil area of 4.57×10^5 km^2 (Liu et al., 2006).

In the second method, the Fe loss was estimated according to the landscape types. It was roughly calculated that the areas of paddy soils developed on sloping uplands, alluvial plains, and bog areas account for 30%, 35%, and 5% of the total paddy soil areas in China, respectively, based on the geographical distribution of paddy soils in China (Li, 1992), the Chinese rice cultivation zoning (China National Rice Research Institute, 1990), temporal and spatial distribution of ancient rice in China (Gong et al., 2007), China's topographic and landform map, and the national soil pH map. Note the areas of paddy soils with shorter cultivation time derived from calcareous parent materials on alluvial plains and bog areas were removed in this calculation because there was no measurable Fe loss for them based on the total Fe density results obtained in this study. According to the difference of Fe density between paddy soils (the average value of paddy profiles with different cultivation age) and non-paddy soil (original soil), and the above-calculated paddy soil areas, we estimated the net Fe losses induced by rice cultivation in different landscapes that were about 4,503.2 Tg for the terraced landscape, 4,204.8 Tg for alluvial plains, and 41.4 Tg for the bog areas, respectively. The net Fe losses from paddy fields in China was about 8,749.4 Tg. The results obtained by the above two methods were very close. We estimated the Fe loss fluxes in paddy soils at the national scale that ranged from 46.4 to 195.7 Tg yr^{-1} based on the data of average Fe loss rates of paddy soils in different

Table 2 Loss flux and loss rate of total Fe in the studied profiles (1 m depth)

No.	Paddy cultivation age (yr)	Loss flux (kg m^{-2})	Loss rate (kg m^{-2} yr^{-1})	Time period (yr)	Loss rate at various time period (kg m^{-2} yr^{-1})
B-C-BY	5	−0.3	−0.056	0–5	−0.056
	20	−1.7	−0.083	5–20	−0.092
	50	−9.7	−0.194	20–50	−0.268
	500	1.8	0.004	50–500	0.026
P-C-CX	50	−7.2	−0.144	0–50	−0.144
	300	−7.4	−0.025	50–300	−0.001
	700	−11.8	−0.017	300–700	−0.011
	1000	−13.3	−0.013	700–1000	−0.005
	2000	15.9	0.008	1000–2000	0.029
P-N-ZS	50	13.6	0.271	0–50	0.271
	200	14.2	0.071	50–200	0.004
	600	51.1	0.085	200–600	0.092
T-N-GH	100	7.7	0.077	0–100	0.077
	1000	14.8	0.015	100–1000	0.008
T-A-YT	30	33.2	1.106	0–30	1.106
	100	47.5	0.475	30–100	0.204
	200	47.8	0.239	100–200	0.003
T-A-JX	60	36.8	0.613	0–60	0.613
	150	39.2	0.261	60–150	0.027
	300	69.1	0.230	150–300	0.200
T-A-LS	30	6.6	0.221	0–30	0.221
	150	25.5	0.170	30–150	0.157
	300	31.4	0.105	150–300	0.039

landscapes and the corresponding paddy soil areas. It should be pointed out that the Fe loss calculations may have certain errors in this method because the Fe density of different landscapes was only based on the paddy soil chronosequences data collected from this study, and the paddy soil area of different landscapes was inferred from indirect data.

In the third method, the total Fe loss in paddy fields at the national scale was estimated based on the data of 351 soil profiles collected from the "Soil Series of China" (Lu, 2017; Ma and Zhang, 2017; Wang, 2017; Zhang et al., 2017; Zhang and Ma, 2017; Wu K N et al., 2019), the national soil parent material map (Xiong, 1986), and the national 100 m land cover data (Zhang, 2007). The results were listed in Table 3. The 351 soil profiles including 134 paddy and 217 non-paddy soil profiles derived from seven types of parent materials. The total area of paddy soils derived from the seven parent materials was 2.07×10^5 km^2. The amount of Fe loss in the paddy fields was 2,319.8 Tg based on this method. Since the area of the paddy soil involved in the calculation of this study was less than 50% of the national paddy soil area, the estimated amount of Fe loss was low. If the paddy soil area was calculated as 4.57×10^5 km^2 (Liu et al., 2006), the total amount of Fe loss from paddy fields in China would be about 5,121.5 Tg. The main sources of error at the national-scale Fe loss from paddy fields estimated with this method included (1) the smaller number of soil profiles that could not cover all parent material types; (2) the relatively high difference of Fe loss amount in different parent materials which induced error when simply restoring the paddy soil area to 100%, and (3) the assumption of a uniform parent material despite the heterogeneity of the same parent material in different regions and even in the same region.

4. Discussion

4.1 Variations and controls of Fe loss during paddy soil evolution

The chronosequence approach has been extensively used in soil genesis research and is a valuable research technique for investigating the rates and directions of pedogenic evolution (Brantley, 2008; Chen et al., 2015). The studied seven sets of paddy soil chronosequences developed from different landscapes (bog, plain, terrace) and parent materials (acidic, neutral, calcareous) with cultivation history from 0 to 2,000 yr presented an excellent opportunity to investigate the spatiotemporal dynamics of soil Fe loss after paddy cultivation and to identify the important factors influencing Fe

Table 3 Fe loss of paddy soils derived from different parent materials[1)]

Parent material	Number of profiles		Soil total Fe density of 1 m depth (kg m^{-2})		Paddy soil area (km^2)	Fe loss (Tg)
	Non-paddy soil	Paddy soil	Non-paddy soil	Paddy soil		
Weathering product of light-colored coarse aplite	50	39	58.4	56.6	35859	64.9
Weathering product of light-colored fine aplite	17	8	65.0	34.0	9023	279.6
Weathering product of sand shale	37	20	71.9	52.3	45983	898.2
Weathering product of limestone	28	12	72.4	61.3	30754	341.2
Quaternary red clay	4	4	52.9	50.4	10406	25.5
Lacustrine sediments	49	36	65.3	60.3	61401	304.0
Coastal sediments	32	15	91.2	62.2	14047	406.4
Total	217	134	–	–	207474	2319.8

1) Soil profile data from "Soil Series of China" (Lu, 2017; Ma and Zhang, 2017; Wang, 2017; Zhang et al., 2017; Zhang and Ma, 2017; Wu K N et al., 2019)

dynamics and possible mechanisms. Our results indicate that the rates and trajectories of Fe evolution showed distinct patterns in the studied seven paddy soil chronosequences. The paddy soils derived from the calcareous lacustrine sediments in the bog area showed a Fe accumulation trend during the earlier stage and then followed by a trend of Fe loss. The Fe evolution in the paddy soils developed on the calcareous marine sediments in the plain area was dominated by the internal cycles of Fe in soil profiles through coupled reducing-eluviation reactions in the surface horizons and oxidation-illuviation in the subsurface horizons within 1,000 yr of paddy cultivation and then followed by Fe net loss. In contrast, Fe in the paddy soils derived from the acidic and neutral parent materials in the plain and terraced upland areas was rapidly lost during the initial stage of paddy cultivation and the amount of Fe loss increased with the increasing paddy cultivation age. Although the Fe loss showed distinct patterns among the studied seven paddy soil chronosequences, net Fe losses of 1 m soil depth would eventually occur at all studied paddy soil chronosequences regardless of the original landscape and parent material.

To better understand the controlling factors of Fe loss during paddy soils evolution, we proposed a model that illustrates the Fe mobilization and translocation pathways in the paddy soil system, soil-water system, and land-sea system and the transformation processes of different phases of Fe in the inter-profile of paddy soils (Figure 4). In soils, Fe forms in various phases, such as primary silicate minerals, pedogenic clay minerals, Fe (oxyhydr) oxides with different degrees of crystallinity, and in organic complexes (Stucki et al., 1988). The weathering of primary silicate minerals in soils releases Fe(II), which is rapidly oxidized in oxic environments and precipitated as poorly crystalline Fe (oxyhydr) oxides. Then the poorly crystalline Fe (oxyhydr) oxides may transform into crystalline Fe (oxyhydr) oxides, form secondary clay minerals, or combine with organic matter to form organic-inorganic complexes (Schwertmann, 1985) during further soil development. Under oxic conditions in well-drained soils, Fe(III) is the thermodynamically stable oxidation state and characterized by a very low solubility. Under anoxic conditions in water-saturated soils, however, Fe(III) provides a terminal electron acceptor for anaerobic respiration of dissimilatory Fe-reducing microorganisms, resulting in the formation of the highly soluble Fe (II) aquoion (Lovley et al., 2004). In addition to reductive dissolution, mobilization of Fe also occurs via proton-promoted dissolution at pH < pznpc (point of zero net protoncharge) and via ligand-promoted dissolution when strong binding organic acids are present (Jansen et al., 2003). As deduced from above information, important factors which influence the transformation and migration of Fe in soil are soil moisture regime and redox condition, organic matter content, soil pH, and soil microbial activities (Schwertmann, 1985; Li X M et al., 2019).

In this study, a trend of net Fe loss occurred in all chronosequences, indicating that the change in soil moisture regime and redox condition caused by artificial submergence and drainage for rice planting was a factor influencing soil Fe loss. The comparison of calcareous and neutral paddy soil chronosequences developed on marine sediment and river alluvium both in the alluvial plain landscape showed significant influences of parent materials on the rates and trajectories of Fe loss. Total Fe loss in the neutral paddy soil chronosequence (P-N-ZS) increased gradually with increasing paddy cultivation age within 600-yr. In contrast, total Fe in the calcareous paddy soil chronosequence (P-C-CX) showed an accumulation trend within 1,000-yr of paddy cultivation. Previous studies have shown that the $CaCO_3$ in soils usually retard the transformation of silicate Fe to Fe oxides (Schwertmann, 1985). The critical redox potentials for Fe reduction and consequent dissolution are between +300 mV and +100 mV at pH 6–7, and −100 mV at pH 8, while at pH 5 appreciable Fe reduction occurs at +300 mV (Gotoh and Patrick, 1974). The $CaCO_3$ content and pH value of original soil in the P-C-CX chronosequence were high (Table 1, $CaCO_3$ was 5.83%, pH was 8.07). The high $CaCO_3$

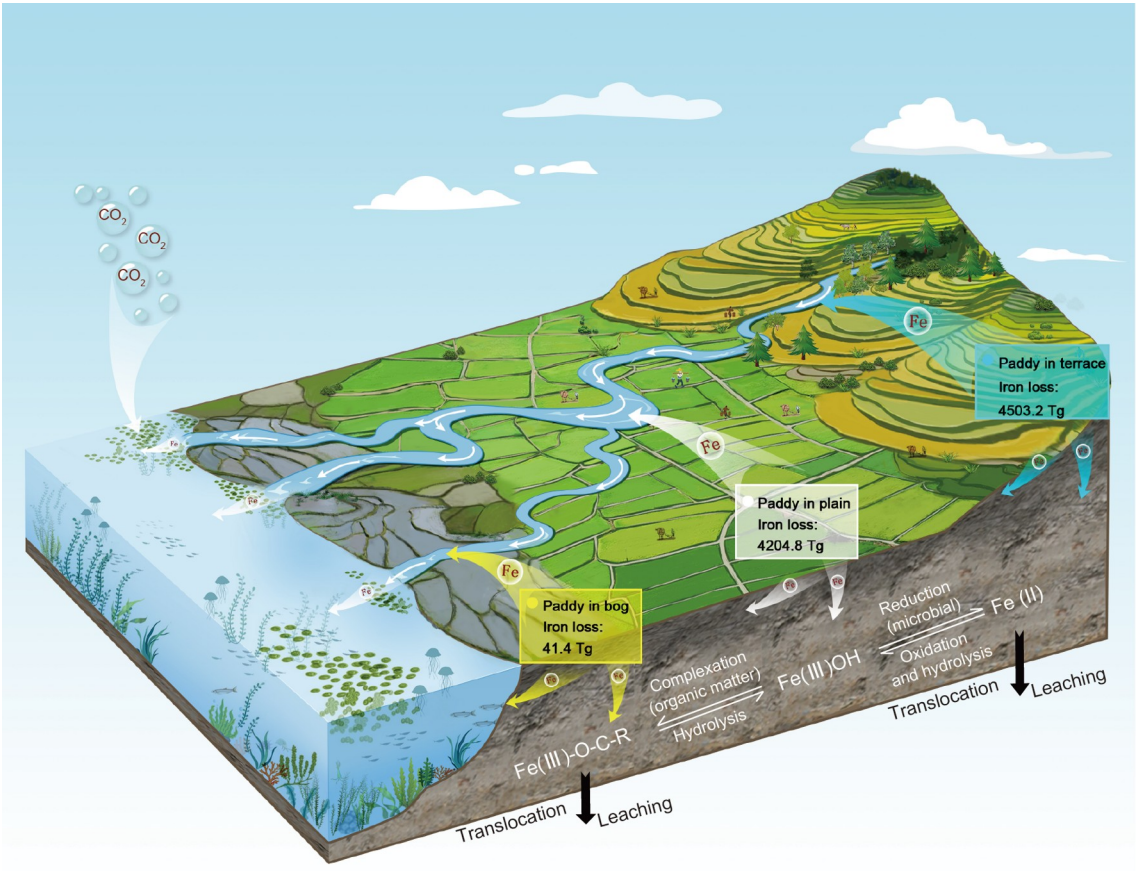


Figure 4 Illustration of Fe transformation and migration paths in paddy soil system, soil-water system, and land-sea system.

content and alkaline soil environment would impede the activation, translocation, and leaching loss of Fe (Gotoh and Patrick, 1974; Schwertmann, 1985). As a result, an initial period of Fe accumulation was observed in the calcareous paddy soil chronosequence. As pedogenesis proceeded and $CaCO_3$ was gradually removed from the profile, the soil pH decreased and net loss of Fe occurred when the cultivation time reached 1,000 yr. Chen et al. (2011) showed that $CaCO_3$ was an intrinsic threshold during paddy soil evolution because profile differentiation of Fe and decreases of magnetic parameter occurred only when carbonates were eventually leached. Therefore, the pH and the content of $CaCO_3$ in the original parent materials were essential factors influencing soil Fe loss. The total Fe stocks in the paddy soil chronosequences developed from calcareous lacustrine sediments in the bog landscape increased with increasing rice cultivation age within 50 yr, which was parallel to the leaching of $CaCO_3$ and decrease of SOC. The results could partly be attributed to the high soil pH and $CaCO_3$ content, which would impede the activation, translocation, and leaching loss of Fe (Gotoh and Patrick, 1974; Schwertmann, 1985). In addition, the increase might be closely related to SOC. Studies have shown that soil organic matter can promote activation, solubility, and migration of soil Fe through reduction and complexation in an anaerobic environment (Jansen et al., 2003; Giannetta et al., 2020). Chi et al. (2021) studied the temporal dynamics in soil Fe affected by wetland conversion on the Sanjiang Plain and the results indicated that soil Fe stock in the 0–120 cm layers gradually increased with increasing planting ages within 18 yr and the decrease in SOC contributed to the changed transportation and dynamics of soil Fe after wetland conversion. In comparison, the acidic environment under an anaerobic condition was conducive to promoting the solubility, mobility, and leaching loss of Fe in the soil (Gotoh and Patrick, 1974; Schwertmann, 1985). In addition to the differences in parent materials, paddy soils developed on the sloping upland areas would have a higher leaching potential than those developed on the bog and plain areas. This was confirmed by the rapid decrease of clay content in the sloping upland paddy soils as compared with the gradual clay enrichment in the paddy soils developed on alluvial plain (Huang et al., 2018a). In addition to the reductive leaching, particle-facilitated leaching and transport of Fe may also explain the rapid decrease of Fe in the acid paddy soil chronosequences developed on the sloping upland areas. Since the external pedogenic environment, human activities, and internal soil properties all changed dynamically over time (Huang et al., 2015), the characteristics of Fe

loss from paddy soils were affected by the length of time for paddy cultivation. To sum up, the loss rates and trajectories of Fe in paddy soils were determined by a combination of internal and external factors, including the pH, $CaCO_3$, and organic matter contents of original soil, paddy soil age, landscape types and positions, and anthropogenic managements.

4.2 Fe loss in paddy fields and its environmental implications

Periodic irrigation of paddy fields promotes soil Fe loss due to the increase of iron mobility after reduction. Zhang (1993) studied the Fe activation, migration and balance within soil compartment based on lysimeters located in Experimental Ecology Station of Red Soil Region and found obvious Fe loss via reduction leaching. Wu H Y et al. (2019) found that conversion from upland to paddy field developed from Quaternary red clay increased the loss of Fe by 44% in the top 3 m within a short timescale (about 20 yr). Chinese and Japanese soil scientists have been paying attention to low-Fe-containing "whitish soil" and "aging soil" (Yu, 1959; Xiong et al., 1980; Kyuma, 2004), and found that the important pedogenic processes of these two types of paddy soil by reductive Mn and Fe eluviation and ferrolysis (Brinkman, 1970). Although soil Fe loss after rice cultivation has been noticed for a long time, quantitative study of Fe loss from paddy fields is limited. A rough estimation of Fe loss in paddy fields at the national scale in this research showed that the Fe loss rate was about 46.4–195.7 Tg yr^{-1}, and the total amount of Fe loss was about 5,121.5–9,412.2 Tg. China has a very long rice cultivation history, with widely rice distribution and various types of paddy soils originated from many types of original soils or different parent materials, and various landscapes. In addition, the Fe cycle in paddy soils is very complex which involves multiple processes such as redox, adsorption, dissolution, precipitation influenced by soil pH, Eh, and microbial characteristic of original soil (Li and Li, 2019), paddy soil age, landscape types and positions, and anthropogenic management. Therefore, the estimation of Fe loss from paddy fields nationwide has great uncertainty. The spatiotemporal dynamic changes of both paddy soil distribution and soil total Fe density may be the main sources of uncertainty. In terms of temporal dynamics, the amount of Fe loss at different paddy cultivation ages was different, but only linear relationships were considered in this study. There might be nonlinear relationships between the amount of Fe loss and the paddy cultivation age. For example, short-term floods or heavy rainfall events may cause sudden changes in the amount of Fe loss in paddy fields. In recent years, the increasing occurrence of extreme climate events (Piao et al., 2019) warrants further study of the relationship between extreme precipitation events and the amount of Fe loss in paddy fields. Spatially, on the one hand, the distribution of different types of paddy soil may change in different regions. The increase or decrease of paddy field area is an important factor influencing the estimation of Fe loss. On the other hand, the paddy soil chronosequences in this study did not involve northern paddy fields. The 351 soil profiles collected from the literature did not cover all parent material types, and there were large differences in the amount of soil Fe loss among paddy soils developed from different parent materials. These factors might have caused errors in the estimation of Fe loss from the paddy soils at the national scale. Therefore, more systematic and comprehensive studies on spatiotemporal dynamic changes of Fe losses from the paddy soils in the future are needed to reduce the uncertainty. Nevertheless, it is currently a more realistic and feasible way to estimate the amount of Fe loss in paddy soils at the national scale using paddy soil chronosequence approach.

In 1990, the famous "Fe hypothesis" was proposed by Martin, a chemical oceanographer (Martin, 1990). He revealed that Fe restricted the productivity of phytoplankton in the HNLC sea area, thereby affecting the transportation of carbon dioxide to the deep ocean. After that, verification of the "Fe hypothesis" and ocean Fe enrichment experiments have become one of the most eye-catching research directions in this field (Zhang and Sun, 2002; Sun et al., 2009; Boyd et al., 2012; Du et al., 2013; Misumi et al., 2014; Rembauville et al., 2018).

Due to the controversy over possible environmental negative effects of the man-made addition of Fe to the ocean, it is particularly important to clarify the contribution of different natural Fe sources to the ocean. The rate and flux of Fe activation and migration in various terrestrial ecosystems vary greatly. Previous research showed that iron supply reached the oceans mainly from rivers as suspended sediment in a vast global transport system (Table 4). However, fluvial particulate iron (625–962 Tg Fe yr^{-1}) and glacial particulate iron (34–211 Tg Fe yr^{-1}) were efficiently trapped in near-coastal areas (Poulton and Raiswell, 2002), except where rivers discharged directly beyond the glacier shelf. Hence, the dominant external input of iron to the surface of the open ocean was aeolian dust transport (16 Tg Fe yr^{-1}), mainly from the great deserts of the world (Jickells et al., 2005). According to the estimates of this study and comparing the various ocean Fe sources and fluxes reported in the literature (Table 5), it was found that the Fe loss from paddy fields might have played an important role in the ocean Fe input. The Fe lost from paddy fields might be in the form of dissolved Fe (Li and Li, 2019), organic-inorganic complexes (Giannetta et al., 2020), and/or colloidal particles (Zhang S G et al., 2019), and the Fe might flow into rivers, lakes, and/or oceans via surface runoff and/or into groundwater via preferential flow (Huang et al., 2018a). The Fe concentrations of suspended solids and

Table 4 Total Fe concentrations in the suspended particles and sediments of the Pearl River, Yangtze River, and Yellow River (Fe_2O_3, wt %)[a)]

Sample source	Suspended particles			Sediments
	Upper stream	Lower stream	Estuary	
Pearl River	–	7.10	–	–
Yangtze River	5.47±1.46	6.43±1.53	7.04	5.49
Yellow River	–	4.03	5.13	3.35

a) Values were calculated based on Yang and Li (1999), Qu et al. (1984), and Poulton and Raiswell (2002)

Table 5 Global Fe fluxes to the ocean

Source	Flux	Reference
Fluvial particulate total Fe[1)]	625–962 Tg yr^{-1}	Poulton and Raiswell (2002)
Glacial sediments[1)]	34–211 Tg yr^{-1}	Poulton and Raiswell (2002)
Atmospheric dust[2)]	16 Tg yr^{-1}	Jickells et al. (2005)
Hydrothermal[2)]	14 Tg yr^{-1}	Poulton and Raiswell (2002)
Coastal erosion[1)]	8 Tg yr^{-1}	Poulton and Raiswell (2002)
Fluvial dissolved Fe[1)]	1.5 Tg yr^{-1}	Poulton and Raiswell (2002)
Dissolved Fe released by Asian glaciers[1)]	23.8 Gg yr^{-1}	Li X Y et al. (2019)
Dissolved Fe released by global glaciers[1)]	185 Gg yr^{-1}	Li X Y et al. (2019)
Fe loss from paddy soil in China[1)]	46.4–195.7 Tg yr^{-1}	This study

1) Partly reach the oceans, 2) all reach the oceans

surface sediments in the Pearl River, Yangtze River, and Yellow River collected from literatures were shown in Table 4 (Qu et al., 1984; Yang and Li, 1999; Poulton and Raiswell, 2002). The total Fe concentrations of suspended solids in water bodies and surface sediments of the Pearl River and the Yangtze River were greater than that of the Yellow River. This might be attributed to the large areas of paddy fields distributed in the Yangtze River and the Pearl River Basin. Because of the differences of the bedrock, weathering environment, and sediment sources of the three river basins, the reasons for the differences in Fe concentrations need to be further evaluated. Ding et al. (2013) systematically studied the characteristics of the chemical composition changes of the suspended particulate materials in different sections of Yangtze River from the source to the estuary. The total Fe data of suspended solids at different times and spaces in this article were collected and were showed in Figure 5. The results showed that the total Fe concentrations of suspended solids in water from the source area to Yibin section (points 1–5 in Figure 5, without paddy fields) were consistently lower than that from Yibin to the estuary section (points 5–25 in Figure 5, with a large area of paddy fields) in different periods (July 2003, July 2005 and July 2007). This may indicate that Fe loss in paddy fields is a potential Fe source of rivers and/or oceans, but this hypothesis needs to be confirmed in further studies. The data obtained in this study could only represent the amounts of Fe loss from the paddy fields. In the future, it is necessary to conduct quantitative research on the form, pathway, and destination of Fe loss from paddy soils. Meanwhile, Fe

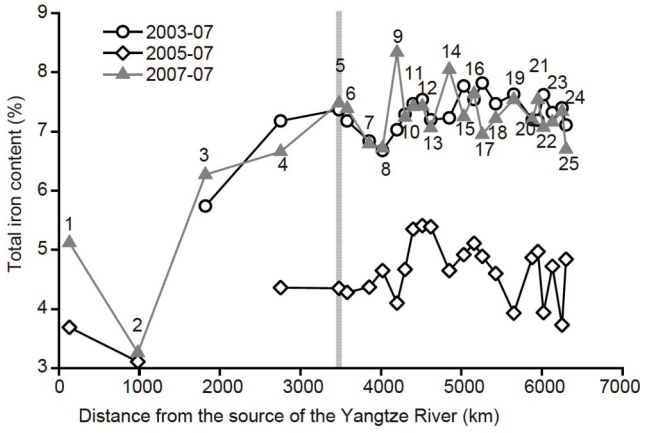


Figure 5 Temporal and spatial variation of total Fe concentration in suspended particulate materials of the Yangtze River. Data from Ding et al. (2013). Sites 1–5 were from the source area to Yibin section, with no paddy fields, and sites 5–25 were from Yibin section to the Yangtze River Estuary, with a large area of paddy fields.

isotope tracing studies (Wiederhold, 2015; Huang et al., 2018b) are suggested to better assess the impact of paddy fields at the global ocean Fe input. Although the Fe loss from soil due to rice cultivation will not directly reach the ocean, the Fe loss fluxes from paddy fields in China is very significant compared to other current Fe loss fluxes (Table 5). At the global scale, a large area of paddy fields is distributed in East Asia, South Asia, South America, and West Africa. The Fe transport flux between land and ocean from these regions may be higher than that of coastal environments without paddy fields, and further systematic eva-

luation is necessary.

5. Conclusion

The studied seven sets of paddy soil chronosequences developed from different landscapes and parent materials over a millennial time scale presented an excellent opportunity to investigate the spatiotemporal dynamics of soil Fe loss induced by paddy cultivation and to identify the factors influencing Fe dynamics. The rates and trajectories of Fe evolution from this study showed distinct patterns among the studied chronosequences. The paddy soils derived from the calcareous lacustrine sediments in the bog area showed a Fe accumulation trend during the 0- to 50-yr time period and then followed by a trend of Fe loss. The Fe evolution in the paddy soils developed on the calcareous marine sediments in the plain area was dominated by the internal cycles of Fe in soil profiles through coupled reducing-eluviation reactions in the surface horizons and oxidation-illuviation in the subsurface horizons within 1,000 yr of paddy cultivation and then followed by Fe net loss. In contrast, Fe in the paddy soils derived from the acidic and neutral parent materials in the plain and terraced upland areas was rapidly lost during the initial stage of paddy cultivation and the amount of Fe loss increased with the increasing paddy cultivation age. The loss rates and trajectories of Fe in paddy soils were determined by a combination of internal and external factors, including the pH, $CaCO_3$, and organic matter contents of original soil, paddy soil age, landscape types and positions, and anthropogenic managements. Based on the available data, it was roughly estimated that the national-scale Fe loss flux from paddy fields was about 46.4–195.7 Tg yr^{-1}, and the amount of Fe loss from paddy fields nationwide was about 5,121.5–9,412.2 Tg. The Fe lost from paddy fields may have a great impact on the biogeochemical cycle of Fe at the regional or global scale. A more systematic and comprehensive study of the temporal and spatial changes of Fe loss in paddy fields, as well as the form, pathway, and destination, will provide an important basis for evaluating the Fe contribution from paddy soils to the ocean.

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